WEEK 12 • IS321 ENTERPRISE ARCHITECTURE

# TOGAF & the Architecture Development Method

*A Full Walkthrough of TOGAF's Core Concepts and All Nine ADM Phases*

TODAY'S ROADMAP

# What We'll Cover

| | | |
|---|---|---|
| 01 | **What Is TOGAF?** | Origins, adoption, and when you'd actually reach for it |
| 02 | **Core Concepts** | BDAT, Building Blocks, the Enterprise Continuum, Governance, Stakeholders & Views |
| 03 | **The ADM: Nine Phases** | Every phase's objectives, approach, inputs, steps, outputs, and artifacts |
| 04 | **Requirements Management** | The hub every other phase revolves around |

# What Is TOGAF?

> *The Open Group Architecture Framework, version 9.1, launched 1 December 2011 — based on the Technical Architecture Framework for Information Management (TAFIM), originally developed by the U.S. Department of Defense.*
>
> — TOGAF's origin

## Two definitions worth knowing

### "Enterprise"

Any collection of organizations with a common set of goals — a government agency, corporation, division, department, or organizations linked by common ownership.

### "Architecture"

A formal description of a system at component level to guide implementation, or the structure of components, their relationships, and the principles governing their design over time.

# Development, Adoption & When to Use It

## Where TOGAF Came From

Developed by more than 300 enterprise architects from leading companies including Dell, Cognizant, and Microsoft. Development traces back to 1995; version 9.1 embodies all improvements made since. It has a common vocabulary and supports all levels of architecture, for enterprises large and small.

**Adopted by 60% of Fortune 500 companies**

*Highly adaptable and collaborative — but criticized for a theory-focused approach few can turn into practice. Use it as a guide, not an actionable plan on its own.*

## When Would You Use TOGAF?

- To transform an organization's architecture and processes in a controlled manner.
- To repeatedly align your processes with current goals.
- To support an architecture development process at all stages — from evaluating key requirements to actual implementation.

CORE CONCEPTS

# Seven Concepts Worth Knowing Cold

*Before the phase-by-phase walkthrough, seven core concepts — "the most important thing on the exam," per TOGAF's own training material:*

- **Four Architecture Domains (BDAT)**
- **Architecture Development Method (ADM)**
- **Deliverables, Artifacts & Building Blocks**
- **Enterprise Continuum**
- **Architecture Repository**
- **Architecture Capability of an Organization**
- **Stakeholders, Views & Viewpoints**

# The Four Architecture Domains (BDAT)



## Business

Business strategy, governance, organization, and key business processes.



## Data

The structure of logical and physical data assets, and data management.



## Application

A blueprint for individual applications, their interactions, and relationships to core business processes.



## Technology

The logical software and hardware capabilities needed to support the above — hardware, networks, etc.

# Deliverables, Artifacts & Building Blocks

### Deliverable

A work product contractually specified, formally reviewed, agreed to, and signed off on by stakeholders — the output of a project.

### Artifact

An architectural work product — a catalog, diagram, or matrix; part of a deliverable; placed into the Architecture Repository.

### Building Block

A potentially reusable component. Architecture Building Blocks (ABBs) are requirements spanning multiple projects; Solution Building Blocks (SBBs) are solutions spanning multiple projects.

# The Enterprise Continuum: Generic to Specific

↓ classifies artifacts in the repository

*"A view of the Architecture Repository that provides methods for classifying architecture and solution artifacts as they evolve from generic to specific." Four tiers, mirrored across two parallel continuums:*

↓ may be asked to classify these tiers

| Tier | What It Means | Architecture Continuum (ABBs) | Solutions Continuum (SBBs) |
|---|---|---|---|
| **Foundation** | Extremely generic — programming languages, operating systems, ITIL | Foundation Architecture | Foundation Solutions |
| **Common Systems** | Still generic, some detail — III-RM, security/network architecture, SaaS vendors like MS Office | Common Systems Architecture | Common Systems Solutions |
| **Industry** | Industry-specific — e.g. credit-card processing in retail, managed by an industry group | Industry Architecture | Industry Solutions |
| **Org-Specific** | Most specific — your business's own services, functions, and implementations | Organization-Specific Architecture | Organization-Specific Solutions |

# Architecture Repository & Architecture Capability

## Architecture Repository

Contains all the building blocks that go into your architecture — essentially an empty box you fill with architectural output as you produce it: deliverables, artifacts, building blocks, and Enterprise Continuum content.

*Compare to Week 5's EA Repository — same idea, TOGAF's vocabulary layered on top.*

## Architecture Capability

↓ key resources and personnel's

"Enterprise Architecture Capability" — the business capability for architecture itself, using organization structures, roles, responsibilities, skills, and processes. Who's the full-time architect? Who do they report to? Is there a governance board? Will everyone actually follow the rules?

# Architecture Governance

*"The practice and orientation by which enterprise architectures... are managed and controlled at an enterprise-wide level. It is concerned with change processes (design governance) and operation of product systems (operational governance)."*

## The Governance Repository

Like the Architecture Repository, but for governance: Reference Data, Process Status, Audit Information, and other documents about the governance process itself.

## Three Levels of Integration

- Strategic architecture (enterprise vision)
- Segment architecture (business-unit / department)
- Capability architecture (specific applications)

# Stakeholders, Views & Viewpoints



## Stakeholder

*"An individual, team, or organization with interests in, or concerns relative to, the outcome of the architecture."*

EXAMPLES: Architect (you), Accounting/Finance, Customer, Business Unit, Partners, IT Group, HR, QA, Security



## View

*"The representation of a related set of concerns — what is seen from a viewpoint." Doesn't have to be visual.*

EXAMPLES: People view, Workflow view, Cost view, Data entity view, Hardware view, Security view



## Viewpoint

*"The perspective from which a view is taken" — the vantage point, not the view itself.*

EXAMPLES: Business stakeholders: cost, value, flexibility. Technical stakeholders: # apps, # servers, uptime.

# Capability Assessment

First created in Phase A, updated in Phase E, and tracked in the Governance Log (alongside the Decision Log, Compliance Assessments, Calendar, Project Portfolio, and Performance Measurement) to ensure progress from baseline to target capabilities.

as-is in TOGAF terminology

to-be

### Business Capability

Baseline and target-state performance, and how each capability is realized.

### IT Capability

Baseline/target maturity of change and operational processes, plus capacity assessment.

### Architecture Maturity

development

Governance processes, roles, skills assessment, and repository/standards quality.

4 Pillars

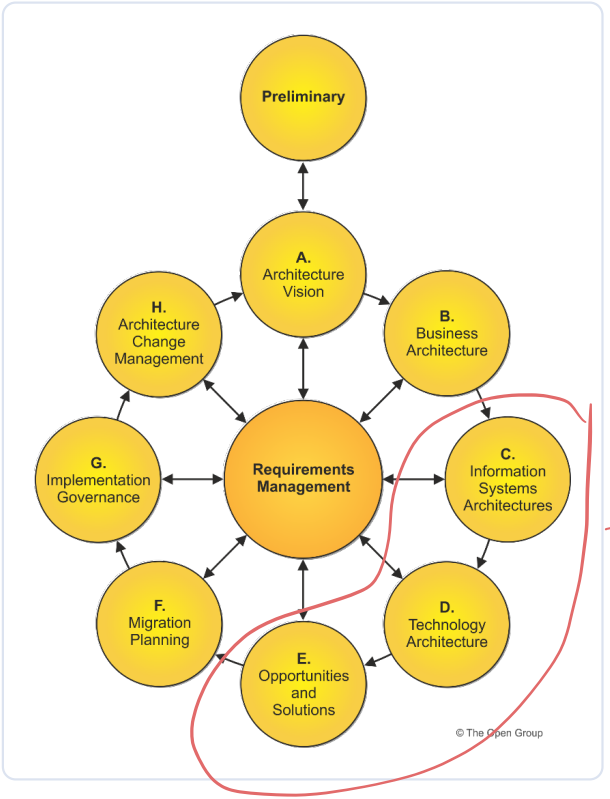
### Business Transformation Readiness

Assessment of how ready the organization actually is to absorb the change.

# The ADM: Nine Phases Around One Hub

"A method for developing and managing the lifecycle of an enterprise architecture" — a preliminary phase, plus eight phases forming a cycle, surrounding the continuously-ongoing Requirements Management phase. Each phase is itself made up of steps.

*Designed to be generic, to fit most enterprises — but can be modified or extended. A smaller company might trim steps it can't support; a company with its own governance framework might substitute that in.*



# The ADM Is Iterative

↓ ✱ distinguish between iterative ADM approaches

### Across the Whole Process

Phase H ends and flows straight back into a new Phase A — the cycle never really stops.

### Between Phases

Phase C work may reveal new business processes, forcing a return to revise Phase B's Business Architecture.

### Within a Phase

It may take several passes to get a detailed target business architecture right before moving on.

**Four dimensions of scope to decide for each iteration:**

- **Breadth of enterprise coverage**
- **Level of detail**
- **Time period for the work**
- **Which BDAT domain(s) to focus on**

# Preliminary Phase



Preparation and initiation activities: defining “the enterprise,” choosing frameworks, evaluating enterprise maturity and capability, and defining architecture principles — everything that happens before Phase A begins.

OBJECTIVES

- Determine the Architecture Capability desired by the organization
- Establish that Architecture Capability

# Preliminary Phase: Approach, Inputs, Steps & Outputs

## Approach

- Define “the Enterprise”
- Identify key drivers & elements
- Define requirements for architecture work
- Define architecture principles
- Define framework to use
- Evaluate EA maturity

## Key Inputs

- TOGAF & other frameworks
- Board strategies & business plans
- IT strategy, business principles/goals/drivers
- Frameworks in use (e.g. Scrum)
- Existing architecture capability
- Partnership & contract agreements

## Key Steps

- Scope enterprise orgs impacted
- Confirm governance/support frameworks
- Define & establish architecture team
- Identify & establish architecture principles
- Tailor TOGAF & other frameworks
- Implement architecture tools

## Key Outputs

- Organization Model for EA
- Tailored Architecture Framework
- Initial Architecture Repository
- Restated business principles/goals
- Request for Architecture Work (optional)
- Architecture governance framework

PHASE 1 OF 9

# Architecture Vision (Phase A)



Develop a high-level vision to be delivered, obtain approval for a Statement of Architecture Work, define scope, estimate resources needed, develop a roadmap and schedule, define KPIs and metrics, and build a communications plan.

OBJECTIVES

- Develop a high-level aspirational vision of the capabilities and business value to be delivered
- Obtain approval for a Statement of Architecture Work

# Phase A: Approach, Inputs, Steps, Outputs & Artifacts

## Approach

- Scope the Request for Architecture Work
- Create the Architecture Vision
- Run the ADM “Business Scenarios” process (use cases)

## Key Inputs

- External reference materials
- Request for Architecture Work
- Business principles/goals/drivers
- Org Model for EA
- Tailored Architecture Framework
- Populated Architecture Repository

## Key Steps

- Establish the architecture project
- Identify stakeholders, concerns & requirements
- Confirm business goals/drivers/constraints
- Evaluate business capabilities
- Assess readiness for transformation
- Develop architecture vision & Statement of Architecture Work

## Key Outputs

- Approved Statement of Architecture Work
- Refined business principles/goals/drivers
- Architecture principles
- Capability assessment
- Architecture vision
- Draft architecture definition doc v0.1
- Communications plan

SOA

## Artifacts

- Stakeholder map matrix
- Value chain diagram
- Solution concept diagram

# Business Architecture (Phase B)

Obtains outputs/deliverables of Phase A to produce a baseline

The "B" in BDAT. Develop the baseline business architecture v1.0 (often bottom-up, based on what exists), develop the target business architecture v1.0 (normally top-down, from strategy), and identify the gaps between the two.

to refine it

## OBJECTIVES

- Develop the target business architecture based on business goals
- Identify candidate Architecture Roadmap components

# Phase B: Approach, Inputs, Steps, Outputs & Artifacts

collect workflows, pain points rolled up to form bigger picture

## Approach

- Business architecture = foundation for other domains
- Develop baseline description bottom-up → operational level
- Use business/activity/use-case models
- Borrow models from industry groups & standards

## Key Inputs

- Business principles/goals/drivers
- Capability assessment & communications plan (Phase A)
- Approved Statement of Architecture Work
- Architecture principles & vision
- Enterprise Continuum & Repository
- Draft architecture definition doc v0.1

## Key Steps

- Select reference models, viewpoints, tools
- Develop baseline business architecture (as-is)
- Develop target business architecture (to-be)
- Perform gap analysis
- Define candidate roadmap components
- Conduct stakeholder review; finalize

## Key Outputs

- Refined Phase A deliverables
- Draft architecture definition doc (baseline+target v1.0)
- Draft architecture requirements spec (gap analysis)
- Business architecture components of roadmap

## Artifacts

- Org/actor & driver/goal/objective catalogs
- Business interaction matrix
- Business footprint diagram
- Business service/function catalog

Phase B output

PHASE 3 OF 9 (DATA)

# Information Systems Architecture — Data

The "D" in BDAT, the first half of Phase C. Data and Application Architecture are developed separately (in either order, or together).
Develop the baseline data architecture v1.0, the target data architecture v1.0, and identify gaps.

↓ similar to Business Architecture

OBJECTIVES

- Develop the target data architecture based on business goals
- Identify candidate Architecture Roadmap components

# Phase C (Data): Approach, Inputs, Steps, Outputs & Artifacts

## Approach

- Address data management issues
- Consider data migration issues
- Consider data governance
- Look for an existing data standard to borrow (e.g. SIF for education, ARTS for retail)

## Key Inputs

- Data principles (Ch. 23)
- Approved Statement of Architecture Work
- Architecture vision & Repository
- Draft architecture docs & requirements (Phase B)
- Business-related architecture roadmap (Phase B)

## Key Steps

- Select reference models, viewpoints, tools
- Develop baseline data architecture
- Develop target data architecture
- Perform gap analysis
- Define candidate roadmap components
- Conduct stakeholder review; finalize

steps for B → C → D are unified w/ different concerns

## Key Outputs

- Refined Phase A deliverables
- Draft architecture definition doc (baseline+target v1.0)
- Draft architecture requirements spec (gap analysis)
- Data architecture components of roadmap

## Artifacts

- Data entity / data component catalog
- Data entity / business function matrix
- Conceptual & logical data diagrams
- Data security, migration & lifecycle diagrams

PHASE 3 OF 9 (APPLICATION)

# Information Systems Architecture — Application



The “A” in BDAT, the second half of Phase C. Develop the baseline application architecture v1.0, the target application architecture v1.0, and identify the gaps between them.

OBJECTIVES

- Develop the target application architecture based on business goals
- Identify candidate Architecture Roadmap components

PHASE C (APPLICATION) — DETAIL

# Phase C (Application): Approach, Inputs, Steps, Outputs & Artifacts

## Approach

- Consult the Architecture Repository
- Look for existing business models to adopt (e.g. TeleManagement Forum for telecom, III-RM from The Open Group)

## Key Inputs

- Application principles (Ch. 23)
- Approved Statement of Architecture Work
- Architecture vision & Repository
- Draft architecture docs & requirements (Phase B)
- Business-related architecture roadmap (Phase B)

## Key Steps

- Select reference models, viewpoints, tools
- Develop baseline application architecture
- Develop target application architecture
- Perform gap analysis
- Define candidate roadmap components
- Conduct stakeholder review; finalize

## Key Outputs

- Refined Phase A deliverables
- Draft architecture definition doc (baseline+target v1.0)
- Draft architecture requirements spec (gap analysis)
- Application architecture components of roadmap

## Artifacts

- Application portfolio & interface catalogs
- Application/organization matrix
- Role/application matrix
- Application Communication diagram

PHASE 4 OF 9

# Technology Architecture (Phase D)



The “T” in BDAT — the final domain-specific phase. Develop the baseline technology architecture v1.0, the target technology architecture v1.0, and identify the gaps between them.

OBJECTIVES

- Develop the target technology architecture based on business goals
- Identify candidate Architecture Roadmap components

# Phase D: Approach, Inputs, Steps, Outputs & Artifacts

## Approach

- Consult the Architecture Repository
- Consider existing IT services
- Use the TOGAF Technical Reference Model (TRM)
- Consider generic industry technical models

## Key Inputs

- Product info on candidate products
- Technology principles (Ch. 23)
- Approved Statement of Architecture Work
- Architecture vision & Repository
- Draft docs from Phases B & C
- Business/data/app roadmap components

## Key Steps

- Select reference models, viewpoints, tools
- Develop baseline technology architecture
- Develop target technology architecture
- Perform gap analysis
- Define candidate roadmap components
- Conduct stakeholder review; finalize

## Key Outputs

- Refined Phase A deliverables
- Draft architecture definition doc (baseline+target v1.0)
- Draft architecture requirements spec (gap analysis)
- Technology architecture components of roadmap

ADD → concludes baseline and target specific to technology

## Artifacts

- Technology Standards & Portfolio catalogs
- Application/Technology matrix
- Environments & Locations diagram
- Platform Decomposition & Processing diagrams

PHASE 5 OF 9

# Opportunities and Solutions (Phase E)

Generate the initial complete version of the Architecture Roadmap based on gap analysis across all of BDAT. Determine whether an incremental approach is needed; formulate an implementation and migration strategy; identify and group work packages.

OBJECTIVES

- Generate the initial complete Architecture Roadmap, based on the BDAT gap analysis
- Determine if an incremental approach is required, and identify Transition Architectures that deliver continuous business value

↓ follows an iterative approach

# Phase E: Approach, Inputs, Steps, Outputs & Artifacts

## Approach

- Figure out how to deliver the target architecture
- Build the Architecture Roadmap
- Define Work Packages
- Define Transition Architectures
- Draft the Implementation & Migration Plan

## Key Inputs

- Product information
- Planning methodologies
- Tailored Architecture Framework
- Approved Statement of Architecture Work
- Draft architecture docs & requirements (BDAT)
- Change requests for existing programs

## Key Steps

- Determine key corporate change attributes
- Review/consolidate gap analysis from Phases B-D
- Reconcile interoperability requirements
- Confirm readiness/risk for transformation
- Formulate implementation & migration strategy
- Identify & group work packages; create the roadmap

## Key Outputs

- Refined Phase A deliverables
- Draft architecture definition & requirements docs
- Capability assessments
- Architecture roadmap v0.1
- Implementation & migration plan v0.1 → will be enhanced to form another version

## Artifacts

- Product context diagram → to study all existing products, their feasibility and ability to deliver solutions
- Benefits diagram → to understand capabilities

PHASE 6 OF 9

# Migration Planning (Phase F)



Finalize the Architecture Roadmap and Implementation & Migration Plan. Ensure the plan is coordinated with the enterprise's approach to change, that stakeholders understand the business value, and start thinking about costs, resources, and timing.

OBJECTIVES

- Finalize the Architecture Roadmap and supporting Implementation & Migration Plan
- Ensure coordination with the enterprise's overall change portfolio
- Ensure the business value is understood by stakeholders

PHASE F — DETAIL

# Phase F: Approach, Inputs, Steps & Outputs

## Approach

- Creation of the Implementation and Migration Plan itself

## Key Inputs

- Governance models
- Tailored Architecture Framework
- Approved Statement of Architecture Work
- Architecture vision & Repository
- Draft architecture docs & requirements (BDAT)
- Change requests for existing programs

## Key Steps

- Confirm management framework interactions
- Assign business value to each work package
- Estimate resource requirements & project timings
- Prioritize
- Confirm architecture roadmap; update definition doc
- Generate the Implementation & Migration Plan

## Key Outputs

- Implementation & Migration Plan v1.0
- Finalized architecture definition document
- Finalized architecture requirements & roadmap
- Reusable ABBs
- Requests for Architecture Work (next ADM cycle, if any)

PHASE 7 OF 9

# Implementation Governance (Phase G)



Ensure conformance with the target architecture. Handle governance duties and change requests from the implementation team, and update the baseline architecture as changes are implemented. The phase ends when solutions are fully deployed.

## OBJECTIVES

- Ensure conformance with the Target Architecture by implementation projects
- Perform appropriate Architecture Governance functions for the solution and any implementation-driven Change Requests

PHASE G — DETAIL

# Phase G: Approach, Inputs, Steps & Outputs

## Approach

- Establish an implementation program to deliver the Transition Architectures
- Adopt a phased deployment schedule reflecting the Architecture Roadmap
- Follow the organization's governance standards
- Use the organization's requirements management process

## Key Inputs

- Architecture Repository & definition documents/requirements (BDAT)
- Architecture roadmap v0.1 (Phase E)
- Implementation governance model
- Architecture contract
- Request for Architecture Work (next cycle, Phase F)

## Key Steps

- Confirm scope and priorities for deployment
- Identify deployment resources & skills
- Guide development of solutions deployment
- Perform Enterprise Architecture Compliance Reviews
- Implement business and IT operations
- Perform post-implementation review; close

## Key Outputs

- Architecture contract (“joint agreements between development partners and sponsors on deliverables, quality, and fitness-for-purpose”)
- Compliance assessments
- Change requests
- Architecture updates

PHASE 8 OF 9

# Architecture Change Management (Phase H)



Keeping the architecture alive. Ensure Architecture Governance is happening and the enterprise's Architecture Capability is maintained — monitoring changes to the business, the industry, technology, and formal change requests.

OBJECTIVES

- Ensure the architecture lifecycle is maintained
- Ensure the Architecture Governance Framework is executed
- Ensure the enterprise Architecture Capability meets current requirements

PHASE H — DETAIL

# Phase H: Approach, Inputs, Steps & Outputs

## Approach

- Drivers for change are not always strategic here -- often reacting to “reality”
- The enterprise change management process determines how changes are managed
- Determine what's simple maintenance vs. what requires redoing the ADM cycle

## Key Inputs

- Statement of Architecture Work & architecture vision (Phase A)
- Architecture definition docs & requirements (BDAT)
- Architecture roadmap (Phase F)
- Change requests (technology, business, lessons learned)
- Architecture contract & compliance assessments (Phase G)

## Key Steps

- Establish the Value Realization Process
- Deploy monitoring tools
- Manage risks
- Provide analysis for Architecture Change Management
- Develop change requirements to meet performance targets
- Manage the governance process; activate change

## Key Outputs

- Architecture updates
- Changes to architecture framework/principles
- New requests for architecture work (major changes)
- Updated Statement of Architecture Work
- Updated architecture contract & compliance assessments

THE CENTER OF EVERYTHING

# Requirements Management: The Hub



Sits at the center of the ADM wheel, operating continuously throughout the whole process rather than being confined to any single phase. Requirements change constantly — a new competitor, new legislation, a disproven assumption, a new company policy — and Requirements Management assesses the impact of each change.

market changes

OBJECTIVES

- Ensure the Requirements Management process is sustained and operates for all relevant ADM phases
- Manage architecture requirements identified during any ADM cycle or phase
- Ensure relevant requirements are available for use by every phase

# Requirements Management: Approach, Inputs & Steps

## Approach

- Continuously drives the ADM cycle from the center
- Each ADM phase identifies which requirements are relevant to it
- No specific tool is recommended within TOGAF for managing requirements

## Key Inputs

- Architecture Repository
- Org Model for EA & Tailored Architecture Framework
- Statement of Architecture Work & Architecture vision (Phase A)
- Architecture requirements
- Requirements impact assessments ★ most common output

## Key Steps

- Identify/document requirements from ADM phases
- Baseline requirements: prioritize, confirm stakeholder buy-in
- Monitor baseline requirements
- Identify changed requirements (add/remove/modify)
- Implement change in the current ADM phase
- Assess & revise gap analysis for past Phases B-D

## Key Outputs

- Requirements impact assessment
- Architecture requirements specification, if necessary

is fed with change request outputs updated specifications

CHAPTER SUMMARY

# Key Takeaways

- TOGAF organizes architecture into four domains (BDAT) and provides a repeatable process — the ADM — for developing it.
- Seven core concepts underpin everything: BDAT, the ADM, Deliverables/Artifacts/Building Blocks, the Enterprise Continuum, the Architecture Repository, Architecture Capability, and Stakeholders/Views/Viewpoints.
- The ADM's 9 phases surround a continuously-active Requirements Management hub, and the whole process is iterative at three levels.
- Phases B, C, and D each follow the same baseline → target → gap-analysis pattern, applied to a different BDAT domain; every phase also has its own Approach, Inputs, Steps, Outputs, and (usually) Artifacts.
- Phases E through H mark the shift from planning the architecture to implementing and governing it.

COMING UP NEXT

# Further Reading & Next Session

## Reading Material

- The Open Group — TOGAF Standard, Version 9.2 (opengroup.org)
- TOGAF 9.1 Specification, Chapters 3, 6–17, 23, 41, 49–50 (definitions, ADM phases, principles, repository, contracts, governance)
- PSU IS321 Syllabus — AI and EA Tools policy

## Next: Week 13

The EA³ Cube Framework, revisited in full, plus AI-Assisted Modeling, AI-Supported Governance, and Smart EA Repositories — tying together Week 3's five levels with TOGAF's process from this week. Mapped to CLO1, CLO3, CLO5.

**Questions?**